

Monotone Blocks and Reverse Replies

An elementary proof of the large-order
permutation-avoidance conjecture

AI-assisted research report
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20 September 2026

Abstract

In the normal-play permutation-avoidance game, a move forbids a length- k pattern that still occurs in a surviving permutation of length n . Ulfarsson’s Conjecture 2.9 asks whether, for each $k \geq 3$ and all sufficiently large n , the second player wins by replying to every pattern with its reversal. We give a self-contained elementary proof with the explicit sufficient bound

$$n \geq R(k) := k + (k + 1)^2(k - 2)^2 + 1.$$

The main ingredient is a minimal-support lemma: in a family consisting of singleton entries and one inflatable monotone interval, deleting singleton entries can isolate exactly one member of any prescribed set of nonmonotone patterns of a common length. A grid pigeonhole argument and the Erdős–Szekeres theorem extract such a family from every sufficiently long legal witness. Reversal then supplies a witness for the required reply at the *same* length n . We also obtain an eventual pattern-isolation theorem, a complement-reply corollary, and explicit small certificates. Exact Python checks and two worked examples accompany the proof; no computation is a premise of the universal result. The bound is not asserted to be optimal.

Status of the result. This is a proposed solution with a complete proof, not a report of an independently accepted theorem. The argument has been internally audited and tested on exact finite examples, but has not been independently refereed or checked in a proof assistant. The source problem is Conjecture 2.9 of arXiv:2603.16004v1, as retrieved on 20 September 2026.

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1 The problem and the main result

1.1 A move removes a pattern, not an entry

Let S_n be the permutations of $\{1, \dots, n\}$. For a sequence of distinct numbers $a_1, \dots, a_j$, its *standardization* $\text{st}(a_1 \cdots a_j)$ is the unique permutation of length j with the same pairwise comparisons. A permutation π contains $p \in S_k$ if

$$\text{st}(\pi(i_1) \cdots \pi(i_k)) = p \quad \text{for some } i_1 < \cdots < i_k.$$

Write $p \leq \pi$ for this relation. Otherwise π avoids p . For a set of patterns F , put

$$\text{Av}_n(F) = \{\pi \in S_n : f \not\leq \pi \text{ for every } f \in F\}.$$

The game studied here is Ulfarsson's PAP game [1]. Its initial position is S_n . When the already forbidden patterns are $F \subseteq S_k$, a legal move chooses $p \in S_k$ occurring in some member of $\text{Av}_n(F)$ and changes the position to $\text{Av}_n(F \cup \{p\})$. The player making the last legal move wins. A selected pattern cannot be selected again, so play lasts at most $k!$ moves.

For $p = p_1 \cdots p_k$, let $p^r = p_k \cdots p_1$. The proposed strategy for Player II is simply

After Player I chooses p , choose p^r .

Although $p \neq p^r$ for $k \geq 2$, that does not by itself make the reply legal: all surviving occurrences of p^r might have disappeared when p was forbidden. Proving that this cannot happen at sufficiently large n is the substantive issue.

1.2 Published target and contribution

Ulfarsson explicitly formulates eventual success of this strategy as Conjecture 2.9 [1]. His paper establishes the $k = 3$ case and the $k = 4$ range $n \geq 10$; the general statement is left open. The following uniform theorem is the proposed resolution.

Theorem 1.1 (Reverse-reply theorem). *For every integer $k \geq 3$, define*

$$R(k) = k + (k + 1)^2(k - 2)^2 + 1 = k^4 - 2k^3 - 3k^2 + 5k + 5. \quad (1)$$

For every $n \geq R(k)$, Player II wins PAP on S_n with pattern length k by always replying with the reversal of Player I's move. In particular, $\text{sg}(S_n, k) = 0$ for every such n .

The proof does not classify the $k!$ patterns individually. It instead minimizes the number of exceptional entries needed to retain *either* orientation of the proposed move. At a minimum, the two orientations cannot both occur. This removes the need for a case-by-case list of companion patterns.

k	3	4	5	6	7	8	9	10
$R(k)$	20	105	330	791	1608	2925	4910	7755

These are sufficient thresholds from one argument, not computed least thresholds. In particular, the displayed $k = 3, 4$ bounds deliberately do not compete with the sharper results already in the source. The new assertion is uniformity in *every* k .

2 Elementary tools and the obstruction to naive pairing

Write

$$\iota_k = 12 \cdots k, \quad \delta_k = k \cdots 21, \quad \mathcal{N}_k = S_k \setminus \{\iota_k, \delta_k\}.$$

The elements of $\mathcal{N}_k$ are the nonmonotone length- k patterns. Define the k -shadow and its nonmonotone part by

$$\text{Sh}_k(\pi) = \{p \in S_k : p \leq \pi\}, \quad \text{Sh}_k^{\text{nm}}(\pi) = \text{Sh}_k(\pi) \cap \mathcal{N}_k.$$

Classical containment is transitive, so if $\rho \leq \pi$ then $\text{Sh}_k(\rho) \subseteq \text{Sh}_k(\pi)$.

Lemma 2.1 (Reversal compatibility). *For all $p \in S_k$ and $\pi \in S_n$,*

$$p \leq \pi \iff p^r \leq \pi^r.$$

Consequently, if $F^r := \{f^r : f \in F\} = F$, then $\text{Av}_n(F)$ is closed under reversal.

Proof. Reverse the order of the selected positions of an occurrence. Their values are then read in reverse order, which standardizes to p^r . Reversal is an involution, giving the converse. Apply the equivalence to each $f \in F$. $\square$

Remark 2.2 (Why symmetry alone is insufficient). The lemma proves that p and p^r are either both legal or both illegal *before* p is played from a reversal-closed state. A legal reply needs more: a surviving permutation containing p^r and avoiding p . Reversing a witness that contains both orientations still contains both and therefore does not solve the problem.

We use the following standard monotone-subsequence theorem, including its elementary proof to keep all dependencies visible. Its classical source is Erdős and Szekeres [2].

Lemma 2.3 (Erdős–Szekeres). *If a permutation has more than $(a-1)(b-1)$ entries, it contains an increasing subsequence of length a or a decreasing subsequence of length b . In particular, more than $(h-1)^2$ entries force a monotone subsequence of length h .*

Proof. At each position j , let u_j and v_j be the lengths of the longest increasing and decreasing subsequences ending there. For $i < j$, if $\pi(i) < \pi(j)$ then $u_j \geq u_i + 1$; otherwise $v_j \geq v_i + 1$. Thus all pairs (u_j, v_j) are distinct. If the two claimed subsequences are absent, all pairs lie in $\{1, \dots, a-1\} \times \{1, \dots, b-1\}$, which has only $(a-1)(b-1)$ elements. $\square$

A *monotone interval* of a permutation is a set of entries occupying consecutive positions and consecutive values, with those entries increasing or decreasing. In the extraction below, the interval property is required only in a selected subpermutation. The corresponding points need not form an interval in the original, much larger permutation.

3 One monotone block and its stable shadow

Definition 3.1 (Pointed template). A pointed template $\mathcal{T} = (\beta, j, \varepsilon)$ consists of a permutation $\beta \in S_{m+1}$, an index j , and a sign $\varepsilon \in \{+, -\}$. For $t \geq 1$, the permutation $\mathcal{T}_t$ is obtained by replacing the entry $\beta(j)$ by a monotone interval of length t and sign ε . Every other entry is retained as a singleton. Thus $|\mathcal{T}_t| = m + t$.

Explicitly, set $v = \beta(j)$. Replace v by

$$v, v+1, \dots, v+t-1 \quad \text{or} \quad v+t-1, \dots, v+1, v,$$

and replace every other value $x > v$ by $x + t - 1$, leaving values $x < v$ unchanged. The sign specifies the choice of order. The block is consequently in one position gap and one value gap relative to all singleton entries.

A subset A of the singleton entries determines a subtemplate $\mathcal{T}^A$: delete the other singleton entries from the skeleton, retain the marked entry, and standardize. For every $t \geq 1$,

$$(\mathcal{T}^A)_t \leq \mathcal{T}_t. \quad (2)$$

Lemma 3.2 (Stable shadows). *For every pointed template and every $k \geq 2$,*

$$\text{Sh}_k(\mathcal{T}_t) = \text{Sh}_k(\mathcal{T}_k) \quad (t \geq k), \quad (3)$$

$$\text{Sh}_k^{\text{nm}}(\mathcal{T}_t) = \text{Sh}_k^{\text{nm}}(\mathcal{T}_{k-1}) \quad (t \geq k - 1). \quad (4)$$

Proof. A k -subsequence chooses some particular singleton entries and a number b of block entries. After the singleton choices and b are fixed, its pattern is fixed: all selected block entries have the same comparisons with every selected singleton, and their internal order is specified by ε . The choice of *which* b block entries is irrelevant.

For the full shadow, $0 \leq b \leq k$, and all choices of b that are possible for $t \geq k$ are already possible for $t = k$. This proves (3). For a nonmonotone pattern, not all k entries can come from the monotone block. Thus at least one singleton is selected and $b \leq k - 1$. All such choices are already available at $t = k - 1$, proving (4). $\square$

Denote this stable nonmonotone shadow by

$$H_k(\mathcal{T}) := \text{Sh}_k^{\text{nm}}(\mathcal{T}_{k-1}).$$

It can be computed from a permutation of length $m + k - 1$, no matter how large the eventual inflated permutation will be. By (2),

$$A \subseteq A' \implies H_k(\mathcal{T}^A) \subseteq H_k(\mathcal{T}^{A'}). \quad (5)$$

Remark 3.3 (The missing monotone pattern is harmless, but not absent). The full shadow need not stabilize at $k - 1$: a block with $k - 1$ entries cannot by itself supply its length- k monotone pattern, whereas a block with k entries can. We will inflate only while all forbidden patterns are nonmonotone. This is exactly why the smaller stabilization length $k - 1$ is valid in the final bound.

4 The minimal-core isolation lemma

The next lemma is the decisive step. It concerns arbitrary sets of targets, not just two reversals.

Lemma 4.1 (Minimal-core isolation). *Let $Q \subseteq \mathcal{N}_k$ be nonempty, and suppose $H_k(\mathcal{T}) \cap Q \neq \emptyset$. There is a subtemplate $\mathcal{U}$ obtained by deleting singleton entries such that*

$$|H_k(\mathcal{U}) \cap Q| = 1. \quad (6)$$

Moreover, $\mathcal{U}$ has at most k singleton entries, and at least one.

Proof. Choose an inclusion-minimal subset A of the singleton entries such that

$$H_k(\mathcal{T}^A) \cap Q \neq \emptyset,$$

and put $\mathcal{U} = \mathcal{T}^A$. Such an A exists because the original finite set of singletons has the required property. Let $m = |A|$.

Consider any occurrence of any member of Q in $\mathcal{U}_{k-1}$. That occurrence must use *every* singleton of A . Indeed, if it omitted a singleton $a \in A$, the same occurrence would survive in $(\mathcal{T}^{A \setminus \{a\}})_{k-1}$, contradicting the minimality of A .

It follows that every occurrence of any target uses exactly the same m singleton entries, and exactly $k - m$ entries of the monotone block. But these choices determine a *unique* relative order: the singleton entries are fixed, all block entries lie in the same position and value gaps relative to them, and the internal order of the block is fixed. Thus all target occurrences standardize to one and the same permutation. Since at least one target occurs, (6) follows.

The existing occurrence uses all m singletons, so $m \leq k$. Also $m \neq 0$: a monotone block alone cannot contain a nonmonotone target. This proves the last assertion. $\square$

Remark 4.2 (What is, and is not, being minimized). We minimize the singleton support needed to retain *some* member of Q . Minimizing the support of one preselected target would not justify excluding the other targets: another target could use a proper subset of that support. Allowing the retained target to change during deletion is essential. In the game application, reversal symmetry subsequently orients the isolated target in the desired direction.

Example 4.3 (The smallest illustrative pair). Let $k = 3$, $Q = \{132, 231\}$, and inflate the first entry of 1342 increasingly. The resulting family is

$$\mathcal{T}_t = 1, 2, \dots, t, t + 2, t + 3, t + 1.$$

For $t \geq 2$, its nonmonotone 3-shadow is $\{132, 231\}$. Delete the singleton $t + 2$ and standardize. The smaller family is

$$\mathcal{U}_t = 1, 2, \dots, t, t + 2, t + 1,$$

whose only nonmonotone 3-pattern is 132. Its reversal has only 231. For instance, at block length two the successive permutations are

$$12453 \longrightarrow 1243 \longrightarrow 3421.$$

The middle and last permutations respectively isolate the two orientations.

4.1 A finite, terminating minimization algorithm

The proof gives a direct procedure. Starting with $\mathcal{T}$, delete any singleton whose deletion leaves at least one target in the stable shadow. Repeat until no deletion is possible. Every successful step reduces the number of singletons by one, so the process terminates. At termination, the same omitted-singleton argument proves that exactly one target remains.

There is no uniqueness assertion for the resulting template or isolated target. Different deletion orders can give different certificates. This freedom is useful: a reply proof needs one suitable witness, not a canonical one.

5 Extracting an inflatable witness from a long permutation

Lemma 5.1 (Same-cell extraction). *Suppose π contains a marked occurrence of $q \in S_k$, and let $h \geq 2$. If*

$$|\pi| \geq k + (k + 1)^2(h - 1)^2 + 1, \tag{7}$$

then π has a subpermutation $\mathcal{T}_h$ formed from that occurrence as k singleton entries and a monotone block of length h in a pointed template. In particular, $q \leq \mathcal{T}_h \leq \pi$.

Proof. Plot the entries as points $(i, \pi(i))$. The k marked positions partition the unmarked positions into $k + 1$ gaps, including the two exterior gaps. Likewise, the k marked values partition the unmarked values into $k + 1$ gaps. The unmarked points therefore lie in $(k + 1)^2$ open cells, indexed by their position gap and value gap.

There are at least $(k + 1)^2(h - 1)^2 + 1$ unmarked points. Some cell contains at least $(h - 1)^2 + 1$ points. Read its values from left to right and apply [theorem 2.3](#); this yields an increasing or decreasing subsequence of length h entirely inside the cell.

Retain exactly those h points and the k marked points, and standardize. The h retained cell points are consecutive in position among the selected points, because no marked point lies inside their position gap. They are also consecutive in value, because no marked value lies inside their value gap. Thus they form a monotone interval relative to the marked singletons. Collapsing this interval to one marked entry gives the required template. $\square$

Corollary 5.2 (Stable extraction). *Let $k \geq 3$ and $|\pi| \geq R(k)$. If $q \in \mathcal{N}_k$ occurs in π , there is a template $\mathcal{T}$ with exactly k singletons such that*

$$q \in H_k(\mathcal{T}) \subseteq \text{Sh}_k^{\text{nm}}(\pi).$$

Consequently, if π avoids $F \subseteq \mathcal{N}_k$, then $\mathcal{T}_t$ avoids F for every $t \geq k - 1$.

Proof. Use $h = k - 1$ in [theorem 5.1](#). The extracted permutation is $\mathcal{T}_{k-1} \leq \pi$, and the original occurrence of q uses only its singletons. The shadow inclusion follows by containment transitivity. Inflating beyond $k - 1$ preserves the nonmonotone shadow by [theorem 3.2](#); all members of F lie in that part of the shadow. $\square$

We can now combine extraction and minimization without mentioning a game.

Theorem 5.3 (Eventual pattern isolation). *Let $k \geq 3$, let $F, Q \subseteq \mathcal{N}_k$, and suppose some $\pi \in \text{Av}_n(F)$ with $n \geq R(k)$ contains a member of Q . There exist a template $\mathcal{U}$ with $1 \leq m \leq k$ singleton entries and a pattern $q_* \in Q$ such that, for every $t \geq k - 1$,*

$$\mathcal{U}_t \in \text{Av}_{m+t}(F), \quad \text{Sh}_k(\mathcal{U}_t) \cap Q = \{q_*\}. \quad (8)$$

In particular, for every $N \geq 2k - 1$ there is a permutation of length N avoiding F and containing exactly one member of Q .

Proof. Choose a member of Q occurring in π and apply [theorem 5.2](#). Its template $\mathcal{T}$ has $H_k(\mathcal{T}) \cap Q \neq \emptyset$ and $H_k(\mathcal{T}) \cap F = \emptyset$. Apply [theorem 4.1](#) to obtain $\mathcal{U}$. Deleting singletons can only shrink the stable shadow, so $\mathcal{U}$ still avoids F . The target intersection is a singleton, say $\{q_*\}$, and stability makes these statements valid for every $t \geq k - 1$.

For the last assertion choose $t = N - m$. Since $m \leq k$ and $N \geq 2k - 1$, we have $t \geq k - 1$. The resulting permutation has exactly N entries. $\square$

Remark 5.4 (The quantifiers matter). The hypothesis requires one witness at a length at least $R(k)$. The conclusion then supplies isolated-target witnesses at every length $N \geq 2k - 1$. It does *not* say that any witness of length $2k - 1$ is already sufficient for the extraction, nor does it assert that a prescribed member of Q can always be isolated. The latter orientation is supplied by symmetry only in the next section.

6 Legal reverse replies and the winning strategy

Theorem 6.1 (Same-length reverse witness). *Let $k \geq 3$, $n \geq R(k)$, and $F \subseteq \mathcal{N}_k$ satisfy $F^r = F$. If $p \in \mathcal{N}_k$ is legal from $\text{Av}_n(F)$, then there exists*

$$\rho \in \text{Av}_n(F \cup \{p\}) \quad \text{such that} \quad p^r \leq \rho.$$

Thus p^r remains legal immediately after p is played.

Proof. A legal move has a witness $\pi \in \text{Av}_n(F)$ containing p . Apply [theorem 5.3](#) with $Q = \{p, p^r\}$, and choose the resulting isolating permutation σ to have length *exactly* n . This choice is permitted because $n \geq R(k) \geq 2k - 1$.

The two targets are distinct: equality $p = p^r$ would force the first and last entries of a permutation of length at least two to agree. If σ contains p^r , it avoids p and is the desired ρ . Otherwise it contains p and avoids p^r . Set $\rho = \sigma^r$. By [theorem 2.1](#), ρ contains p^r , avoids p , and still avoids F because $F^r = F$. $\square$

For completeness, the monotone endgame and the passage from replies to a winning strategy are proved explicitly rather than imported.

Lemma 6.2 (The monotone endgame). *If $F \subseteq \mathcal{N}_k$ and $n \geq k$, both ι_k and δ_k are legal from $\text{Av}_n(F)$. After one is played, the other remains legal. If moreover $n \geq (k - 1)^2 + 1$, forbidding both leaves no surviving permutation.*

Proof. The increasing permutation ι_n and decreasing permutation δ_n avoid every nonmonotone pattern. They therefore survive F and witness the two monotone moves. If ι_k is chosen, δ_n still survives and contains δ_k ; the opposite case is identical with the orders reversed. The final assertion is [theorem 2.3](#) with $a = b = k$. $\square$

Proof of [theorem 1.1](#). At the beginning of a round, just before Player I moves, maintain the following invariant: either the game has ended on Player II's previous move, or the forbidden set F is reversal-closed and contains no monotone pattern. The invariant holds initially because $F = \emptyset$.

If Player I chooses a nonmonotone p , [theorem 6.1](#) makes p^r a legal reply. Adding both p and p^r preserves reversal-closure and introduces neither monotone pattern, so the invariant is restored.

If Player I chooses a monotone pattern, [theorem 6.2](#) makes its reversal a legal reply. After that reply no permutation survives, since $n \geq R(k) \geq (k - 1)^2 + 1$. Thus Player II wins immediately.

There is no possibility of an infinite run consisting only of nonmonotone rounds. Each round forbids two distinct new patterns, and there are only $k! - 2$ nonmonotone patterns. Before a monotone round, both monotone moves remain legal. Therefore play reaches a monotone round after finitely many rounds, and Player II makes the last legal move.

In a finite impartial normal-play game, a position is losing for the next player exactly when its Sprague–Grundy value is zero; a proof of this standard equivalence is included in [section C](#). Hence $\text{sg}(S_n, k) = 0$ as asserted. $\square$

6.1 Stronger local conclusion

The proof applies not only to the initial state S_n , but to every state $\text{Av}_n(F)$ with $F \subseteq \mathcal{N}_k$ and $F^r = F$. From any such state, the next player loses against reversal replies by the other player when $n \geq R(k)$. No reachability assumption on F is needed.

It is important that the forbidden set at a round boundary contains no monotone pattern. Once Player I first chooses one, we do not invoke the inflation argument again: the very next reverse reply finishes the game.

7 A fully explicit length-330 certificate

The following example exercises the first length-5 bound in [theorem 1.1](#). It is illustrative, not an additional assumption. Let $p = 12354$, so $p^r = 45321$. Take the skeleton

$$\beta = 124635$$

and inflate its fifth entry, of value 3, decreasingly. There are five singleton entries, so block length $t = 325$ gives

$$\pi = 1, 2, 328, 330, 327, 326, \dots, 3, 329 \in S_{330}. \quad (9)$$

The singleton entries themselves, at positions 1, 2, 3, 4, 330, form p . The stable nonmonotone 5-shadow of this template is

$$H = \{12354, 12435, 12453, 12534, 12543, 13524, 14325, \\ 14532, 15324, 15432, 35214, 43215, 45321, 53214\}.$$

It contains both p and p^r . Define the reversal-closed forbidden set

$$F = \mathcal{N}_5 \setminus (H \cup H^r).$$

Here $|F| = 92$. All members of F are avoided by (9).

Minimization with $Q = \{12354, 45321\}$ deletes three singleton entries. The skeletons, with the marked entry underlined and its block always *decreasing*, are

$$1246\underline{3}5 \longrightarrow 135\underline{2}4 \longrightarrow 24\underline{1}3 \longrightarrow 23\underline{1}. \quad (10)$$

At every step at least one target remains. The final template has two singletons and, for every $s \geq 4$, the permutation

$$\mathcal{U}_s = s + 1, s + 2, s, s - 1, \dots, 1.$$

Its only nonmonotone 5-pattern is 45321: selecting both singletons and three block entries yields that pattern, while using at most one singleton yields a decreasing pattern. Thus the length-330 reply witness is

$$\rho = 329, 330, 328, 327, \dots, 1. \quad (11)$$

It contains 45321, avoids 12354, and avoids all 92 patterns of F . No final reversal happens to be needed in this example.

The file `data/worked_certificate.json` records the full input and output permutations, the exact forbidden set, the selected cell, the monotone subblock, every deletion, and the final stable shadow. All of its pattern assertions can be checked on representatives with a block of length four or six; enumerating the $\binom{330}{5}$ subsequences of the large permutations is unnecessary.

8 Constructive algorithm and certificate size

8.1 Input and output

For a nonmonotone move, the constructive input consists of k , n , a reversal-closed $F \subseteq \mathcal{N}_k$, and a witness $\pi \in \text{Av}_n(F)$ together with one marked occurrence of p . Supplying the occurrence avoids a separate pattern-search problem. The output is an explicit $\rho \in S_n$ witnessing the legality of p^r after p .

Reverse-witness construction

1. Split the unmarked entries of π into the $(k+1)^2$ cells defined by the marked occurrence. Choose a cell with at least $(k-2)^2 + 1$ entries.
 2. Find an increasing or decreasing subsequence of length $k-1$ in that cell. Retain it together with the marked occurrence, producing $\mathcal{T}_{k-1}$.
 3. Repeatedly delete a singleton whenever the stable shadow still meets $Q = \{p, p^r\}$. Let $\mathcal{U}$ be the resulting template with m singletons.
 4. Form $\sigma = \mathcal{U}_{n-m}$. If its isolated target is p^r , output σ ; otherwise output σ^r .
-

8.2 Finite proof certificates

A certificate can store the original marked occurrence, the selected $k-1$ cell points, the template and deletion trace, the isolated orientation, and the final block length. The initial finite representative has $2k-1$ entries, and every later representative has at most that many.

The verifier checks that the original selected points standardize to the stated template; checks finite pattern shadows of the representatives; checks that the final shadow contains exactly one target and avoids F ; and, when necessary, invokes reversal-closure of F . The mathematical stability lemma then transports these finite checks to the length- n output. Thus the certificate is independent of n except for the indices of selected points and the final block length.

More precisely, apart from the supplied F and permutation, it contains $O(k^2)$ small skeleton entries if the whole deletion trace is retained, and $O(k \log n)$ bits for the selected original indices. Retaining only the initial and final templates reduces the structural record further, though the full trace is easier to inspect.

8.3 A conservative running-time bound

With a marked occurrence supplied, gap assignment takes $O(nk)$ elementary comparisons. A straightforward dynamic program finds the required monotone subsequence in $O(n^2)$ time, or faster methods can be substituted without changing the proof. Each finite shadow test enumerates at most $\binom{2k-1}{k}$ subsequences; a direct standardization takes $O(k^2)$ comparisons. At most $k(k+1)$ deletion attempts suffice for a naive repeated scan. For fixed k , the overall construction is therefore polynomial in n ; a conservative bound is

$$O\left(n^2 + k^4 \binom{2k-1}{k}\right)$$

comparison operations, apart from representing and testing membership in F . There is no assertion of polynomial dependence on k .

The supplied implementation instead computes stable shadows symbolically: it chooses the singleton subset and only the *number* of selected block entries. An independent direct subsequence routine checks these symbolic computations in the test suite.

9 Extensions and the limits of the argument

9.1 Complement replies

For $p \in S_k$, define p^c by $(p^c)_i = k+1 - p_i$. Complementation preserves classical containment, has no fixed permutation when $k \geq 2$, and exchanges the two monotone patterns.

Corollary 9.1 (Complement strategy). *For every $k \geq 3$ and $n \geq R(k)$, Player II also wins PAP on S_n by replying to p with p^c .*

Proof. At a round boundary keep F complement-closed and free of monotone patterns. Apply [theorem 5.3](#) to $Q = \{p, p^c\}$. If necessary, complement the isolated witness to orient it toward p^c . It still avoids F . The two monotone patterns are paired by complementation, so [theorem 6.2](#) and the same termination argument apply. $\square$

This is a direct consequence of the isolation mechanism, not a separate claim that a new game has been solved. The mechanism requires a symmetry with no fixed target and a suitable endgame. For example, reversal followed by complementation can fix patterns, so this argument does not automatically give a reply strategy for that operation.

9.2 Forbidden patterns of bounded, unequal lengths

The targets must have a common length in [theorem 4.1](#), but the forbidden patterns need not. The distinction permits the following useful extension.

Corollary 9.2 (Bounded forbidden lengths). *Let $k \geq 3$, let $Q \subseteq \mathcal{N}_k$, and let F consist of nonmonotone patterns of lengths at most ℓ . Set $K = \max(k, \ell)$. If some F -avoiding permutation π containing a member of Q has length at least*

$$k + (k + 1)^2(K - 2)^2 + 1,$$

then for every $N \geq k + K - 1$ there is an F -avoiding permutation of length N containing exactly one member of Q .

Proof. Extract a same-cell monotone block of length $K - 1$ around a marked target occurrence. The nonmonotone shadow of every length $j \leq K$ is already stable at that block length. Hence enlarging the block preserves avoidance of every member of F . Minimize singleton support while retaining a target of length k . The proof of [theorem 4.1](#) is unchanged, and leaves at most k singletons. Inflating to total length $N \geq k + K - 1$ leaves at least $K - 1$ block entries, preserving all required avoidance conditions. $\square$

This is a statement about permutation classes and witnesses. It does not settle a game allowing moves of different lengths: the pairing and endgame would require a separate analysis.

9.3 Why the main hypotheses are substantive

The proof has four boundaries. First, the stable representative must be long enough for every relevant nonmonotone pattern; replacing $k - 1$ by a smaller number without a separate argument could introduce new forbidden patterns during inflation. Second, every member of Q has the same length; otherwise using all the singletons need not force the same number of block entries. Third, avoiding monotone forbidden patterns is not preserved by unbounded monotone inflation. Fourth, reversal-closure of F is what permits us to orient the isolated target without losing the previous avoidance conditions.

None of these conditions is silently assumed away. They hold exactly where they are used in the PAP proof.

10 Exact computations and reproducibility

The universal proof consists of [theorems 3.2, 4.1](#) and [5.1](#) and the game argument. The following computations were carried out as independent finite consistency checks and as an aid to discovery.

10.1 Two shadow implementations

The function `shadow_direct` enumerates actual index subsets of a finite permutation and standardizes their values. The function `Template.stable_shadow` instead enumerates singleton subsets and the number of block entries. These use different enumeration schemes, although both necessarily use elementary rank standardization.

For every skeleton in S_{k+1} , every marked slot, and both block signs, the two routines were compared at block lengths $k-1, k, k+1$ for $k = 3, 4$. The resulting exhaustive counts are as follows.

k	Templates	Reverse-pair minimizations	Other target-set tests
3	192	404	2192
4	1200	6482	0

For each reverse-pair minimization, the final target intersection and its minimality under every singleton deletion were checked by the direct routine. At $k = 3$, all nonempty target subsets were also tested whenever present in the starting template.

There were additionally 120 deterministically sampled templates for each of $k = 5, 6, 7$, with comparisons at block lengths $k-1$ and $k+1$, and minimizations for target sets containing up to four sampled patterns and their reversals. These are sampled tests, not exhaustive tests in those three lengths.

10.2 Constructed replies and extraction tests

For each $k = 3, 4, 5, 6, 7$, twenty full length- $R(k)$ replies were constructed from one-block witnesses, using the maximal reversal-closed forbidden set disjoint from each input shadow. This gives 100 certified large-permutation outputs. Their final compressed representatives were checked directly for avoidance and the required target orientation.

A separate set of tests applied same-cell extraction to shuffled, unstructured permutations with empty forbidden set. The suite also checked the sharp $(h-1)^2$ monotone-subsequence obstruction for $2 \leq h \leq 9$, and twenty shuffled sequences of length $(h-1)^2 + 1$ for each of those values.

Every assertion in the executed suite passed. The test seed is 20260920 and is explicitly separate from the operating-system randomness used to choose the research area. The exact output is in `data/verification.json`; the complete length-330 example is in `data/worked_certificate.json`.

10.3 The discovery computation

Before the minimal-core lemma was recognized, a finite exploration enumerated one-block shadows with exactly k singletons. A reversal support records which nonmonotone reversal pairs meet a shadow, without recording their orientations. For each support S arising in the enumeration and every pair in S , the program looked for a separating template whose support was contained in S .

k	Templates	Distinct full shadows	Supports	Failed supports
5	8640	7336	3502	0
6	70560	61816	30794	0

Those tests suggested a support-minimization principle, but by themselves would not prove the all- k theorem. The later argument permits deleting singletons and handles every k without

these tables. The package retains a reproducible exploration script and summary output so that the discovery record is distinguished from the proof dependencies.

10.4 Running the package

The main verification requires Python 3.10 or later and no third-party packages. From the archive root, run

```
python code/verify.py
python code/explore_supports.py --k 5
python code/explore_supports.py --k 6
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The exploration is optional. The first command checks the examples and implementation; the last builds this article from its source. No network connection is required to run the checks or build the PDF once the ordinary L^AT_EX packages are installed.

11 What remains to be determined

The existence assertion in the selected conjecture is the part addressed by [theorem 1.1](#). The argument leaves the quantitative threshold largely unoptimized. The quartic expression has a transparent origin: k marked entries create $(k + 1)^2$ cells, and a cell with more than $(k - 2)^2$ points contains the required block. This is an upper bound on the worst-case extraction cost, not evidence that quartic growth is necessary for the game.

One concrete next question is the least eventual reverse-strategy threshold for $k = 5$. Our result bounds it above by 330, but neither the proof nor the recorded computation determines it. More generally, improving the use of the marked occurrence or exploiting restrictions imposed by F might lower the uniform bound.

The report does not compute the full Sprague–Grundy function, analyze optimal play lengths, prove a misère analogue, or handle arbitrary nonclassical pattern notions. The construction is tailored to classical subsequence containment and normal play. These limitations do not leave a missing case of Conjecture 2.9: they concern different or stronger questions.

A A compact proof audit

The following checklist isolates the places where an incorrect proof of a pairing strategy could otherwise hide a quantifier or length error.

Potential gap	Where it is closed
A reverse witness still contains the just-forbidden pattern.	Minimal-core isolation gives <i>exactly one</i> of p, p^r before any symmetry is applied.
The long monotone block is not an interval in the original permutation.	Only the selected $2k - 1$ points are retained; being in one cell makes the block an interval in that selected subpermutation.
Inflating a witness creates a forbidden pattern.	Every forbidden pattern at this stage is nonmonotone of length k , so its occurrence is already visible at block length $k - 1$.
The minimized occurrence uses a different singleton set.	Minimality forces <i>every</i> target occurrence to use all retained singletons, not merely one chosen occurrence.

Potential gap	Where it is closed
Different block choices give different target patterns. The isolated target is the wrong orientation.	The block is monotone and in one position/value gap; fixing the number chosen and all singletons fixes every relative comparison. Reverse the witness; $F^r = F$ preserves prior avoidance.
The witness has the wrong length.	With $m \leq k$ singleton entries, choose block length $n - m \geq k - 1$ to obtain exactly n entries.
A monotone move violates the inflation hypotheses.	Such a move is handled only by the separate monotone endgame, which ends the game on the very next reply.
The reply rule does not imply eventual termination.	Every legal move forbids a new one of the $k!$ patterns. Before a monotone round, both monotone moves remain legal.
A finite test is treated as a universal proof.	No assertion in the main proof depends on the computational tables or on an unverified classification.

B Selection, exclusion, and source-status record

The research area was selected by one call to Python’s `secrets.randbelow(96)` after fixing a list of 96 mathematical areas. The recorded zero-based result was 9, hence entry 10: *Permutation patterns*. The draw was recorded at 2026-09-20T21:58:38.955311+00:00. The complete list and output are in `area_selection.json`, and `select_area.py` refuses to overwrite an existing draw. The record is an audit trail of the executed choice, not a cryptographic proof of when a human first saw the list.

The entire attached `manifest(1).tex` was read, including its final continuation. It contains no entry for PAP, reverse-reply strategies, or Ulfarsson’s Conjecture 2.9. The manifest’s permutation-counting and permutation-avoidance entries concern different problems; none is selected here. A hash of the supplied manifest and the exclusion rationale are recorded in `notes/selection_and_exclusions.md`.

One initial candidate from the stack-sorting literature was rejected after finding a later paper proving it [3]. The selected source was then checked directly in its versioned arXiv HTML and PDF. Conjecture 2.9 appears on PDF page 5, and the general- k discussion appears in Section 6.1. Targeted searches by paper identifier, title, and conjecture wording did not identify a later resolution. This is a documented source check, not a complete bibliographic guarantee that no independent or unpublished proof exists.

The contribution claimed here is the elementary proof of the stated general reverse-reply theorem through same-cell extraction and minimal-core isolation. Classical monotone-subsequence arguments, the PAP rules, and the previous small- k results are not claimed as new. The precise priority of the abstract isolation lemma beyond this application has not been settled by an exhaustive literature search.

C Why the Sprague–Grundy conclusion follows

For a finite impartial normal-play game, define the Sprague–Grundy value recursively by

$$\text{sg}(X) = \text{mex}\{\text{sg}(Y) : Y \text{ is a legal follower of } X\},$$

where the minimal excludant is the least nonnegative integer not in the set. A terminal position has value zero. Induct on the maximum remaining play length. If $\text{sg}(X) = 0$, every follower has nonzero value and, by induction, is winning for the next player; therefore X is losing for its next player. If $\text{sg}(X) > 0$, zero must occur among the follower values, so the next player can move to

a losing position and win. This proves the equivalence used in [theorem 1.1](#). PAP is finite because each selected pattern becomes permanently illegal, so the induction applies.

References

- [1] Henning Ulfarsson, *A Permutation Avoidance Game with Reverse Replies and Monotone Traps*, arXiv:2603.16004v1, 2026. Conjecture 2.9; Section 6.1. <https://arxiv.org/abs/2603.16004v1>. Versioned HTML and PDF consulted 20 September 2026.
- [2] P. Erdős and G. Szekeres, *A combinatorial problem in geometry*, Compositio Mathematica **2** (1935), 463–470. https://www.numdam.org/item/CM_1935__2__463_0/.
- [3] Yang Li, Sergey Kitaev, Zhicong Lin, and Jing Liu, *A bijection between 321- and 213-avoiding permutations preserving t -stack-sortability*, arXiv:2507.09187, 2025. <https://arxiv.org/abs/2507.09187>. Used only to exclude an already-resolved candidate, not as a proof dependency.